

# JANARDHANA REDDY M S

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## Education

### New York University

Master of Science in Computer Science

Sep. 2024 – May 2026

New York City, New York

### Relevant Coursework

Software Engineering, Algorithms, Big Data, Machine Learning, Cloud and Big Data, Natural Language Processing

### Toyota Kirloskar, Inc

Dec 2023 – Mar 2024

SDE Intern

Bangalore, KA

- Built a logistics mishap dashboard using **Django, PostgreSQL, and REST APIs**, reducing issue resolution time by **30%**.
- Automated invoice approval for a logistics inventory system processing **100,000+** parts and orders, shrinking processing time by **35%** and decreasing error rates by **15%**.
- Enhanced the data processing pipeline by implementing data validation checks, reducing manual effort by **20 hours/week**.

### Robolution

Mar 2022 – July 2023

Vice President of Club - Software

Bangalore Institute of Technology

- Led a 10 member software team to build and deploy a **fine-tuned YOLOv6 object detection model** on industry-specific warehouse datasets, achieving **92% mAP** for autonomous navigation of mini robots in industrial environments.
- Optimized model inference for embedded hardware deployment using **low-level C++ systems programming**, debugging **GCC memory ordering issues** in multi-threaded control systems managing concurrent sensor input, motor commands, and vision processing.
- Led team to win the **Grid Hackathon**, competing against university teams across India with a fully integrated autonomous warehouse robot system combining deep learning perception and real-time embedded control.

## Projects

### Project Cerebro: BCI Motor Imagery System | PyTorch, TensorRT, Kafka, PySpark, MLFlow, Go, LangGraph May 2026

- Architected a **distributed BCI pipeline** that reads brain signals non-invasively from paralyzed patients and predicts which direction they intend to move, ingesting **2048 Hz multi channel EEG** via Kafka → MNE → Spark within **500ms latency** across a polyglot store of **Cassandra, MongoDB, Redis & pgvector**.
- Engineered a **cross-hardware transfer learning ensemble** (EEGNet + EEG Conformer + ShallowConvNet, Platt scaled) pre-trained on **140,000+** labeled trials across 162 subjects, achieving **78.9% macro F1** on held-out cross-hardware test data versus **46.7% zero-shot baseline**.
- Trained the ensemble across **8 A100 nodes** using **PyTorch DDP** with **torchrun**, batch size 64 per GPU and **NCCL** backend for gradient synchronization, reducing training time to support rapid cross-hardware iteration.
- Optimized PyTorch ensemble inference using **TensorRT FP16** on A100 GPU, reducing inference latency from **250ms to 40–50ms**, an **80%+ reduction** over the baseline FastAPI PyTorch serving pipeline.
- Built an **automated MLFlow evaluation gate** that blocked model promotions failing cross-hardware macro F1 thresholds before reaching the serving layer, catching **two bad models** that passed single-device validation but failed on unseen hardware.

### Code Sensei | Go, TypeScript, React, AWS EKS, ECS, DynamoDB, LangGraph, PyTorch

May 2025

- Built an **agentic AI powered interview platform** with two specialized LLM agents, an Assessment Agent and a Hint Agent each with **stateless architecture** persisting all session state externally in **DynamoDB and Redis**, enabling any pod to handle any request with full auditability across **1M+ concurrent users** on AWS EKS.
- Implemented **progressive hint level tracking** per question to constrain the hint agent across in three levels from conceptual nudges to pseudocode, combined with **controlled context retrieval** loading only current question, latest code submission, execution result, and prior hints per request, preventing solution leakage while **minimizing prompt token usage**.
- Built **isolated code execution environments** using **ECS Fargate** with CPU and memory caps, network restrictions, and prebuilt language containers, tested adversarially with prompt injection attacks including role bypass and session manipulation until the model consistently redirected users toward reasoning rather than revealing solutions.

### Crowd Monitoring and Social Distance Detection | PyTorch, YOLOv8, OpenCV, Python

Nov 2024

- Fine-tuned a **YOLOPose v8 object detection model** on a custom annotated crowd dataset with human bias, achieving **92.1% accuracy** for person detection and keypoint localization in dense, crowded outdoor scenes.
- Built an **automated homography pipeline** using OpenCV and pinhole project camera model to transform 2D pixel coordinates from YOLOPose keypoint outputs into real world spatial coordinates, enabling accurate distance estimation and color coded crowd density mapping across thousands of camera feeds.
- Published findings as an **IEEE paper** at NMITCON 2024 (DOI: 10.1109/NMITCON62075.2024.10698968), with the system providing real time awareness of the situation and density alerts for public safety authorities.

## Technical Skills

**ML & AI:** PyTorch, TensorFlow, scikit-learn, TensorRT, MLFlow, LangGraph, CUDA, LoRA Fine-tuning, Platt Scaling

**Languages:** Python, Go, C++, TypeScript, JavaScript, SQL

**Data & Infrastructure:** Apache Kafka, Apache Spark, Delta Lake, Kubernetes, Docker, PostgreSQL, Cassandra, pgvector, Redis

**Cloud:** AWS (EC2, S3, ECS, EKS, Lambda, CloudWatch), Terraform

**Concepts:** Distributed Training, Model Evaluation, Transfer Learning, RAG, Agentic Architecture, CI/CD